

[RI/2024/Q4]

4 Name your Jupyter Notebook as:

TASK4_<your name>_<class>_<index number>.ipynb

A clinic management system maintains records of patients, medical staff and appointments in a database `CLINIC.db`, which is provided with this question.

The database file contains the following tables:

- `Patient (PatientID, Name, DOB, ContactNumber)`: Stores information about patients, including their unique patient ID, name, date of birth, and contact number.
- `Staff (StaffID, Name, Role, Specialization)`: Stores information about medical staff, including their unique staff ID, name, role (Doctor/Nurse/Technician), and specialization.
- `Appointment (AppointmentID, PatientID, StaffID, AppointmentDate, Diagnosis)`: Records appointments made by patients, including the appointment ID, patient ID, staff ID, appointment date, and diagnosis.

For each of the sub-tasks 4.1 to 4.3, add a comment statement at the beginning of the code using the hash symbol '#', to indicate the sub-task the program code belongs to, for example:

```
In [1]: #Task 4.1
        Program code
```

Output:

Task 4.1

The text files `PATIENT.txt`, `STAFF.txt` and `APPOINTMENT.txt` store the comma-separated values for each of the tables in the database.

Write a Python program to read in the data from each file and then store each item of data in the correct place in the database.

Run your program. [4]

Task 4.2

Write an SQL query to retrieve the following data: *Appointment ID, Patient ID, Patient Name, Staff ID, Staff Name, Appointment Date, Diagnosis*

In a new cell, write a Python program that uses this SQL query to retrieve the data and display them.

Run your program. [5]

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Task 4.3

Write an SQL query to:

- Count the total number of appointments each staff member (Doctor/Nurse/Technician) has handled.
- Group the results by role and order them by the number of appointments in descending order.
- Display the staff members' names, roles, and the number of appointments they have handled.

In a new cell, write a Python program that uses this SQL query and display the data under suitable headings. [6]

Save your Jupyter Notebook for Task 4.

Task 4.4

For this sub-task, save your program separately with a .py file extension.

Write a Python program and the necessary files to create a web application. The web application offers the following menu options:

1. Search Appointment
 2. List Staff Workload

For menu option 1, design a web form that allows users to search for appointments by patient name and appointment date. The form should:

- Include input fields for *patient name* and *appointment date*.
- Have a submit button labeled "Search".

The results of the search function should be shown on a web page in a structured table format that shows all details from the Appointment table, including patient name and staff name. If no appointments are found, display a message "No appointments found".

Test your program by selecting menu option 1, and inputting 'David Tan' as patient name and '13/08/2024' as date.

For menu option 2, display the data retrieved from the SQL query in sub-task 4.3, under these headings on a web page: Staff Name, Staff Role, Number of Appointments handled.

[RI/2024/Q4 Cont ']

Save your program code as:

TASK4_4_<your name>_<class>_<index number>.py

with any additional files/subfolders as needed in a folder named

TASK4_4_<your name>_<class>_<index number>

Run the web application and save the output as

- TASK4_4_<your name>_<class>_<index number>_1.html when menu option 1 is selected, input 'David Tan' as patient name and '13/08/2024' as date.
- TASK4_4_<your name>_<class>_<index number>_2.html when menu option 2 is selected.

[14]